WEEKLY REPORT 2

CSE623: Machine Learning Theory And Practice

Crop Yield Prediction Using Classical Machine Learning Models and Climatic Factors

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The Visionary

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1 Introduction

During this week, the focus was on preparing the collected dataset for machine learning analysis through data cleaning, preprocessing, and exploratory data analysis (EDA). These steps are essential for improving data quality and ensuring that machine learning models perform effectively.

2 Work Done This Week

The following tasks were completed during Week 2:

- Importing the agricultural dataset into the Python environment.
- Inspecting the dataset structure, including number of records, attributes, and data types.
- Identifying missing values and duplicate records in the dataset.
- Cleaning the dataset by removing duplicates and handling missing values.
- Converting categorical variables such as crop type, state, and season into numerical format using encoding techniques.
- Performing preliminary Exploratory Data Analysis (EDA) to understand the distribution of key variables such as production, area, and yield.
- Generating visualizations using Matplotlib to observe relationships between important features.

3 Data Preprocessing

Data preprocessing was carried out to ensure that the dataset is suitable for machine learning algorithms.

3.1 Handling Missing Values

Missing values were checked across all attributes. Any incomplete entries were either removed or handled appropriately to maintain dataset consistency.

3.2 Duplicate Removal

Duplicate records were identified using dataset inspection methods and removed to prevent biased model learning.

3.3 Encoding Categorical Variables

Several attributes such as crop type, state, and season were categorical in nature. These variables were converted into numerical representations using encoding techniques so that machine learning algorithms can process them effectively.

3.4 Feature Preparation

Relevant features such as cultivated area, climatic conditions, and crop attributes were prepared for further analysis and modeling.

4 Exploratory Data Analysis

Exploratory Data Analysis was conducted to understand the underlying structure and patterns present in the dataset. Key observations from the analysis include:

- Distribution of crop production across different states.
- Relationship between cultivated area and crop yield.
- Variations in production across different seasons.
- Identification of potential outliers in production and yield values.

Visualizations such as histograms, bar charts, and scatter plots were generated to better understand the dataset characteristics.

5 Work to be Done Next Week

The following tasks are planned for Week 3:

- Splitting the dataset into training and testing sets.
- Implementing regression models such as Linear Regression.
- Implementing ensemble models such as Random Forest and Gradient Boosting.
- Training and evaluating models using appropriate performance metrics.
- Comparing model performance to identify the most accurate approach for crop yield prediction.